

Lab 6: Logistic Regression

DUE: 13:00h 12/06/2026

Submission: mlearning

- **Lab#_YOURNAME.ipynb**
- Must include all data file (.csv,...) and .ipynb in your submission
- .csv and .ipynb must be in the same folder (pd.read_csv('data.csv'))
- No use of LOCAL libraries/folders
- NO LATE submission accepted

Input: creditcard.csv

1. Compute the number of fraud vs genuine Transactions in the dataset.
2. Compute the percentage of fraud vs genuine Transactions in the dataset.
3. Use **Logistic Regression**, compute the accuracy, recall, precision, F1.
Use train_size = 0.70, test_size = 0.30
4. You might notice that the dataset is **unbalanced**. Now **undersample** the genuine class so that its sampled size matches the fraud sample size. Now retrain the model and recompute accuracy, recall, precision, F1. Use train_size = 0.70, test_size = 0.30
5. From your observations, how did undersampling affect the performance?

